

MH1101 Tutorial 5 (Week 6)

Reference: Sections 3.2, 3.3 (Lecture Notes)

1. Evaluate the following trigonometric integrals:

(i) $\int \sin^6 x \cos^3 x \, dx$

(ii) $\int_0^\pi \cos^4(2t) \, dt$

(iii) $\int \tan x \sec^3 x \, dx$

(iv) $\int \tan^2 x \sec^4 x \, dx$

(v) $\int \sin 3x \sin 6x \, dx$

(vi) $\int_0^{\pi/4} \tan^4 t \, dt$

(vii) $\int \tan^2 x \cos^3 x \, dx$

2. Evaluate the integral.

(i) $\int \frac{1}{t^2 \sqrt{t^2 - 16}} \, dt$

(ii) $\int_0^1 \frac{1}{(x^2 + 1)^2} \, dx$

(iii) $\int \sqrt{5 + 4x - x^2} \, dx$

Hint: Express $\sqrt{5 + 4x - x^2}$ as $\sqrt{a^2 - (x - b)^2}$.

(iv) $\int_0^1 \sqrt{x - x^2} \, dx$

Hint: Express $\sqrt{x - x^2}$ as $\sqrt{a^2 - (x - b)^2}$.

(v) $\int \frac{1}{\sqrt{x^2 + 2x + 5}} \, dx$

Hint: Express $\sqrt{x^2 + 2x + 5}$ as $\sqrt{(x + b)^2 + a^2}$.

3. Evaluate the integral

$$\int \tan^2 x \sec x \, dx.$$

Answer Keys. **1.** (i) $\frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + C$ (ii) $\frac{3\pi}{8}$ (iii) $\frac{\sec^3 x}{3} + C$ (iv) $\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$ (v) $\frac{\sin 3x}{6} - \frac{\sin 9x}{18} + C$ (vi) $\frac{\pi}{4} - \frac{2}{3}$ (vii) $\frac{\sin^3 x}{3} + C$ **2.** (i) $\frac{\sqrt{t^2-16}}{16t} + C$ (ii) $\frac{\pi}{8} + \frac{1}{4}$ (iii) $\frac{9}{2} \sin^{-1} \left(\frac{x-2}{3} \right) + \frac{x-2}{2} \cdot \sqrt{5+4x-x^2} + C$ (iv) $\frac{\pi}{8}$ (v) $\ln \left| \frac{\sqrt{x^2+2x+5}}{2} + \frac{x+1}{2} \right| + C$ **3.** $\frac{1}{2} (\tan x \sec x - \ln |\sec x + \tan x|) + C$ □